THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET B

Chapters 3 & 4: **The World of Numbers, Exploring Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A rational number in lowest terms has denominator $2^2 \times 5^4$. Its decimal expansion terminates after how many places? 1

(a) 6 **(b)** 4 **(c)** 3 **(d)** 5

2. The value of $(-20) \div 4$ is: 1

(a) 5 **(b)** 4 **(c)** -3 **(d)** -5

3. Rational numbers are NOT closed under: 1

(a) division by zero (undefined) **(b)** multiplication
(c) subtraction **(d)** addition

4. The absolute value $|-\frac{3}{5}|$ is: 1

(a) $-\frac{3}{5}$ **(b)** 0 **(c)** $\frac{5}{3}$ **(d)** $\frac{3}{5}$

5. Which of the following has a non-terminating repeating decimal expansion? 1

(a) $\frac{3}{20}$ **(b)** $\frac{1}{8}$ **(c)** $\frac{1}{7}$ **(d)** $\frac{7}{25}$

6. Factorise $x^2 - x - 42$: 1

(a) $(x-7)(x-6)$ **(b)** $(x+6)(x+7)$ **(c)** $(x-7)(x+6)$ **(d)** $(x-6)(x+7)$

7. $(3s+5t)(3s-5t)$ equals: 1

(a) $9s^2 - 25t^2$ **(b)** $9s^2 + 30st + 25t^2$ **(c)** $9s^2 - 25t^2 + 4$ **(d)** $9s^2 - 30st + 25t^2$

8. The side of a cube of volume $p^3 + 6p^2q + 12pq^2 + 8q^3$ is: 1

(a) $p+q$ **(b)** $p+2q$ **(c)** $2p+q$ **(d)** $p+4q$

9. Which of the following is an identity? 1

(a) $(x+y)^2 = x^2 + 2xy + y^2$ **(b)** $2x = 10$ **(c)** $x + 5 = 9$ **(d)** $x^2 - 1 = 24$

10. A student writes $(a+b)^2 = a^2 + b^2$. The correct response is: 1

(a) Correct only for $a > b$ **(b)** Correct for all positive a, b
(c) Incorrect; the term $2ab$ is missing **(d)** Correct for all a, b

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. The temperature in Ladakh is 4 °C at noon and drops by 15 °C by midnight. Find the midnight temperature. 2

12. Show that $\frac{4}{6}$ and $\frac{10}{15}$ are equal rational numbers. 2

13. Factorise $x^2 + 11x + 30$ by splitting the middle term. 2

14. Factorise $x^2 - 5x + 6$ by splitting the middle term. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the rational number x such that $\frac{x}{3} + \frac{x}{2} = \frac{16}{3}$. 3

16. A square has side $(2x + 3)$ units. Find its area and perimeter. 3

17. Show that $\sqrt{2}$ can be located on the number line and describe the construction. 3

OR

Classify as rational or irrational: 3

(i) $\sqrt{81}$

(ii) $\sqrt{12}$

(iii) $0.\overline{3}$

(iv) $0.12345\,12345\ldots$ (repeating block). Give the fraction where rational.

18. Is $(a + b - c)^2 + (a - b + c)^2 + (a - b - c)^2 = 2a^2 + 2b^2 + 2c^2$ an identity? Test with $a = 1, b = 1, c = 1$. 3

OR

Saira arranges a square of side x, 8 strips $x \times 1$ and 15 unit squares into a rectangle. Find its length and breadth. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Convert into $\frac{p}{q}$: 5

(i) 12.6

(ii) 0.0120 (terminating)

(iii) 3.052 (terminating)

(iv) $0.\overline{23}$

(v) $2.0\overline{5}$.

OR

(i) Show that between any two rational numbers there is another rational number. 5

(ii) Find the rational number midway between $-\frac{1}{4}$ and $\frac{5}{12}$ and place it on the number line (describe).

20. **Case study:** A cube has edge $(a + b)$. 5

(i) Split it into 2 cubes and 6 cuboids and state their volumes.

(ii) Write the identity.

(iii) If $a = 2$, $b = 3$, verify the identity numerically.

OR

Prove that $(x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx) = x^3 + y^3 + z^3 - 3xyz$ and evaluate $1^3 + 2^3 + (-3)^3$. 5

END OF QUESTION PAPER — Total: 40 Marks

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